Unified Guided Hyperspectral Image Denoising by Continuous Coupled Tucker Decomposition

Xiaoxuan Ji, Pengxian Li, Jialin Wang, Shuang Xu, Teng-Yu Ji, Jiangjun Peng, *Member, IEEE*, Xiangyong Cao, *Member, IEEE*, and Deyu Meng, *Member, IEEE*

***Abstract*—Hyperspectral image (HSI) denoising is a critical preprocessing step for subsequent interpretation tasks. While guided denoising utilizing auxiliary high-quality images [e.g., panchromatic (PAN), multispectral (MSI), or synthetic aperture radar (SAR)] has shown promise, existing methods are modality-specific and fail to generalize across different guidance types. This article proposes a unified guided HSI denoising framework termed continuous coupled Tucker decomposition (CCTD). The proposed method performs a joint factorization of the HSI and guidance modalities via Tucker decomposition, sharing the spatial factor matrices and core tensor across modalities to implicitly align spatial structures while preserving modality-specific spectral characteristics through independent spectral factors. To further exploit internal spatial smoothness priors without the burden of cumbersome hyperparameter tuning, the factor matrices are parameterized as continuous functions of coordinates via implicit neural representations (INRs) with sine activations, which naturally encode smoothness through architectural inductive bias. An explicit nuclear norm regularizer on the spectral mode eliminates the need for manual Tucker rank selection. Extensive experiments on three datasets (Beijing, Wuhan, and Florence) with six noise cases and three guidance modalities (MSI, SAR, and PAN) demonstrate that CCTD generally outperforms state-of-the-art single-image and guided denoising methods, achieving superior peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), erreur relative globale adimensionnelle de synthése (ERGAS), and spectral angle mapper (SAM) metrics. Ablation studies validate the effectiveness of each component. The proposed framework provides a versatile solution for multimodal HSI restoration.**

***Index Terms*—Guided denoising, hyperspectral image (HSI) denoising, Tucker decomposition.**

Received 21 April 2026; revised 26 May 2026; accepted 26 June 2026. Date of publication 29 June 2026; date of current version 8 July 2026. This work was supported in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515011851 and Grant 2023A1515011589, in part by the National Natural Science Foundation of China under Grant 12571571 and Grant 62272375, in part by Sichuan Science and Technology Program under Grant 2026NSFSC0785, in part by the Young Talent Fund of Xi'an Association for Science and Technology under Grant 0959202513207, in part by the Natural Science Basic Research Program of Shaanxi under Grant 2025JC-YBQN-061, and in part by the Practice and Innovation Funds for Graduate Students of Northwestern Polytechnical University under Grant PF2026038. *(Corresponding author: Shuang Xu.)*

Xiaoxuan Ji, Pengxian Li, and Teng-Yu Ji are with the School of Mathematics and Statistics, Northwestern Polytechnical University, Xi'an 710129, China.

Jialin Wang is with the Changqing Oilfield New Energy Institute, PetroChina, Xi'an 710018, China.

Shuang Xu and Jiangjun Peng are with Research and Development Institute of Northwestern Polytechnical University, Shenzhen 518063, China, also with the Advanced Power Research Institute of Northwestern Polytechnical University, Chengdu 610213, China, and also with the School of Mathematics and Statistics, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: xs@nwpu.edu.cn).

Xiangyong Cao is with the School of Electronic and Information Engineering, Xi'an Jiaotong University, Xi'an 710049, China.

Deyu Meng is with the School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an 710049, China.

Digital Object Identifier 10.1109/TGRS.2026.3708294

I. Introduction

HYPERSPECTRAL images (HSIs) capture rich spectral information across hundreds of narrow bands, enabling precise material identification and analysis, with applications in land cover classification [1], environmental monitoring [2], and medical diagnostics [3]. However, due to sensor limitations and atmospheric interference, HSIs are often corrupted by mixed noise (e.g., Gaussian, stripe, and impulse), which severely degrades data quality and hinders downstream tasks such as classification and target detection [4]. Consequently, HSI denoising has become an essential preprocessing step to recover clean signals while preserving both spatial and spectral fidelity [5].

Over the past few decades, significant progress has been made in HSI denoising. Early methods, such as wavelet transform and spatial filtering, demonstrated moderate success but often struggled to retain fine details while adequately removing noise [6], [7]. This limitation accelerated the development of more advanced techniques based on internal prior modeling, which exploit inherent properties of HSI data—such as low-rank property [8], [9], tensor sparsity [10], spatial–spectral smoothness [e.g., total variation (TV)] [11], [12], and nonlocal self-similarity [13]. Recent works further combine these priors with tensor decomposition [14], [15] or deep learning [16], [17] to enhance robustness. Despite their effectiveness under moderate noise, these single-HSI methods face a fundamental bottleneck: when noise levels are high, the internal priors themselves become unreliable because the noisy observations provide insufficient clean signal information [18]. For instance, low-rank models may fail when noise disrupts the intrinsic tensor structure, and deep learning approaches require large labeled datasets that are often unavailable [19], [20], [21].

The inherent limitation of single-image denoising comes from its reliance on information extracted from the corrupted target itself. To overcome this limitation, a promising alternative is *guided denoising*, which leverages complementary information from coregistered high-quality images, such as panchromatic (PAN), multispectral (MSI), or synthetic aperture radar (SAR) data. As illustrated in Fig. 1, these guidance modalities typically share similar spatial structures with the

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

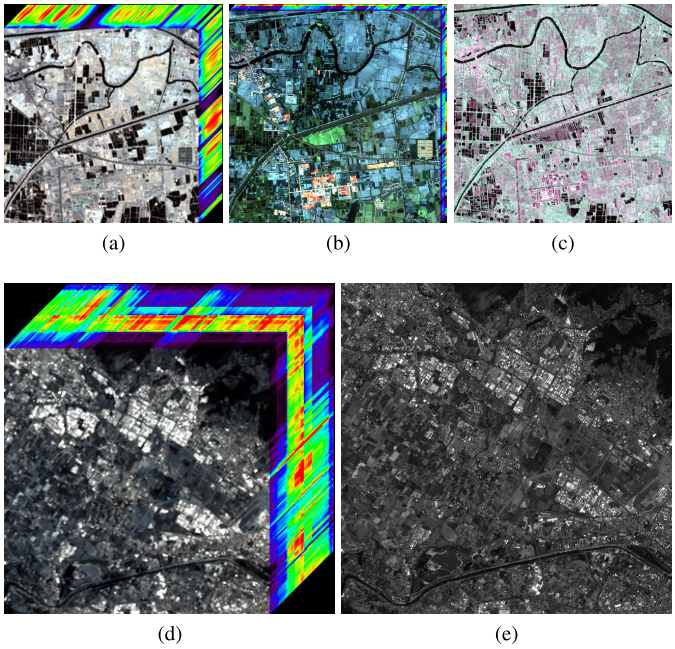


Fig. 1. First row: paired multimodal images of the Wuhan dataset. Second row: paired multimodal images of the Florence dataset. (a) Wuhan-HSI. (b) Wuhan-MSI. (c) Wuhan-SAR. (d) Florence-HSI. (e) Florence-PAN.

target HSI but exhibit lower noise levels, thereby providing a valuable external prior that helps distinguish signal from noise more reliably. This paradigm offers the potential for superior performance, particularly for severely degraded HSIs where internal priors are insufficient [14], [22].

Multimodal data have been widely explored in HSI-related tasks for many years. Early research primarily investigated the fusion of hyperspectral, MSI, and PAN images [23], [24] to support downstream applications, such as feature extraction [25], [26], classification [27], and object detection [28]. By comparison, the use of auxiliary multimodal data for guided HSI denoising has only recently emerged as a new research direction. One of the most notable works in guided HSI denoising is the pan-denoising method [18], which specifically exploits PAN imagery as a guidance source. PAN images capture broad spectral information and share strong spatial correlations with HSIs, making them a natural candidate for providing external structural cues. The method in [18] introduces a PAN-weighted representation coefficient TV (PWRCTV) that uses gradient maps from the PAN image to assign adaptive weights (that is, larger in smooth areas and smaller near edges), thereby preserving edges by aligning gradients between the PAN and HSI. While PWRCTV demonstrates promising results, its applicability is strictly limited to PAN–HSI pairs. This limitation is twofold. First, it requires the guidance modalities to be single-channel, so it cannot directly handle multimodal data such as MSI or SAR. Second, even if one were to average all channels of an MSI or SAR image to obtain a single-channel proxy, the effectiveness would be questionable due to the fundamental differences in imaging mechanism, and therefore, the spatial coherence assumption (guidance modalities and HSI textures are strongly correlated) may not hold for MSI or SAR guidance, severely limiting the PWRCTV's generalizability.

Accordingly, there is a distinct lack of a universal framework capable of effectively integrating various guidance modalities for HSI denoising. The core challenge resides in bridging the considerable cross-modal discrepancy between the HSI and the guidance modalities in an effective manner. To tackle this challenge, a unified guided denoising method is proposed in this article, called continuous coupled Tucker decomposition (CCTD), which is grounded in a shared continuous representation and implemented by combining implicit neural representations (INRs) [29], [30] with Tucker decomposition, as shown in Fig. 2. Although INR and Tucker decomposition have been extensively explored in previous HSI denoising studies, most existing methods mainly treat them as approaches for modeling denoising priors and still lack a unified mechanism for simultaneously addressing cross-modal discrepancies and modality-specific characteristics. To fill this gap, the proposed CCTD integrates Tucker decomposition and INR into a unified continuous coupled framework for multimodal guided HSI denoising. In particular, the HSI and the guidance modalities are decomposed into corresponding factor matrices, with two spatial factor matrices shared across the two modalities. Such a design achieves implicit alignment of cross-modal spatial structures while retaining the spectral uniqueness of each modality, thereby enabling flexible and effective guidance from PAN, MSI, or SAR. Furthermore, to mitigate the inherent subjectivity associated with rank selection in Tucker decomposition, a low-rank regularization term is introduced into the reconstructed HSI, effectively decoupling rank determination from the optimization procedure. In summary, this article presents the following key contributions.

1) The proposed CCTD framework serves as a pioneering approach that leverages shared continuous representations through Tucker-parameterized INRs to bridge modality discrepancies in guided HSI denoising.
2) To reduce the subjectivity of rank selection in Tucker decomposition, a low-rank regularization is incorporated into the reconstructed HSI, ensuring a more robust and automated reconstruction framework.
3) Extensive experiments on simulated and real datasets demonstrate that CCTD generally outperforms state-of-the-art approaches, offering a versatile and powerful solution for multimodal HSI denoising.

The remainder of this article is organized as follows. Section II reviews related work. Section III details the CCTD framework. Section IV presents experimental results and analysis. Section V provides discussions and limitations of the method.

II. Related Work

A. Single HSI Denoising

Single HSI denoising methods have been extensively researched, aiming to enhance the quality of HSIs by removing noise while preserving essential information. Early approaches primarily relied on spatial or transform domain techniques. These methods, while simple, often struggled to balance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

noise removal and detail preservation effectively. Recent optimization-based methods mainly focus on loss functions and leveraging internal image priors.

1) Loss Functions: The ℓ_2 loss is very sensitive to noise, whereas the ℓ_1 loss is more robust because it measures the sum of absolute differences, making it less influenced by extreme values. Therefore, ℓ_1 loss is a more popular choice. However, from the maximum likelihood principle, ℓ_1 loss assumes the noise is sampled from the Laplacian distribution. However, it is challenging to verify whether the noise comes from the Laplacian distribution. A mixture of Gaussians is often used to approximate complex statistical distributions, particularly when the number of Gaussian components is large enough to capture the distribution's characteristics. Applying this concept to hyperspectral denoising, mixtures of Gaussian or exponential power distributions are often used to model unknown and non-i.i.d. noise. Recent studies have shown that real-world hyperspectral noise often exhibits asymmetric distribution, and explicitly modeling this asymmetry can lead to improved denoising performance [31].

2) Internal Priors: Frequently used priors for single HSI denoising include low-rank, local smoothness, nonlocal self-similarity, and hybrid priors. The low-rank prior assumes the HSI lies in a low-dimensional manifold, and can be imposed through regularizations (such as nuclear norm, tensor nuclear norm, and their variations) [32], [33] or matrix/tensor decomposition models [34]. The local smoothness prior often incorporates TV regularization into objective functions to enforce spatial coherence. On the other hand, the nonlocal self-similarity prior assumes that similar image patches or structures can be found at different locations within an HSI, enabling information aggregation across the entire image to enhance denoising performance [35], [36].

These individual priors have their own strengths and limitations, and their effectiveness can vary depending on the specific characteristics of the HSI data and the nature of the noise [37]. While these priors can be incorporated into a single objective function to potentially boost performance, doing so often requires laborious hyperparameter tuning. To address this challenge, very recent research has focused on developing hybrid priors that aim to combine multiple priors into a unified regularization framework. For example, by observing the low-rank prior for gradient maps, correlated TV (CTV) [38] is defined as the nuclear norm of gradient maps. This could model both low-rank and local smoothness priors in CTV. Another typical example is the representative coefficient TV (RCTV) [39]. After applying matrix factorization, the representative coefficient matrix could retain the image prior. Imposing TV regularization on the representative coefficient could achieve a similar aim.

B. Guided HSI Denoising

Under conditions of severe noise corruption, the internal priors inherent in HSIs often become unreliable. Consequently, recent research has shifted toward integrating external priors from auxiliary images (e.g., PAN, MSI, or SAR), a paradigm known as guided denoising. Existing guided HSI denoising methods mainly exploit auxiliary information through either prior-based regularization or multimodal tensor modeling.

For prior-based regularization, the pioneering PWRCTV method [18] uses PAN images to adaptively weight TV regularization, assigning smaller weights to high-gradient regions (edges) to preserve them and larger weights to smooth areas to suppress noise. This strategy effectively preserves spatial details while suppressing noise, outperforming single-HSI methods. However, PWRCTV is designed for single-channel PAN guidance and not easily adapted to other modalities for guidance.

For multichannel guidance, DTucker [40] separately decomposes HSI and MSI and transfers multimodal information by imposing a mapping constraint between their core tensors. Such a constraint may weaken guidance details, thereby limiting its ability to provide precise guidance for HSI denoising.

Therefore, despite the progress of existing guided denoising methods, it remains challenging to develop a unified framework that can flexibly and precisely exploit different guidance modalities for HSI denoising.

C. Preliminaries

This section commences with the definition of essential tensor operations that form the theoretical foundation of the proposed framework. A tensor is a multidimensional array, with an N-way tensor as $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$. Its mode-$n$ unfolding $\mathbf{X}_{(n)} \in \mathbb{R}^{I_n \times (I_1 \cdots I_{n-1} I_{n+1} \cdots I_N)}$ arranges all mode-n fibers to be the columns of the resulting matrix. More precisely, the element $(i_1, i_2, \ldots, i_N)$ of the original tensor maps to the matrix element (i_n, j) in $\mathbf{X}_{(n)}$, where

$$j = 1 + \sum_{\substack{k=1 \\ k \neq n}}^{N} (i_k - 1) J_k, \quad \text{with} \quad J_k = \prod_{\substack{m=1 \\ m \neq n}}^{k-1} I_m.$$

The mode-n product of a tensor $\mathcal{X}$ with a matrix $\mathbf{U} \in \mathbb{R}^{J \times I_n}$ yields a new tensor $\mathcal{Y} = \mathcal{X} \times_n \mathbf{U}$ of size $I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_N$, with elements defined as

$$(\mathcal{X} \times_n \mathbf{U})_{i_1, \ldots, i_{n-1}, j, i_{n+1}, \ldots, i_N} = \sum_{i_n=1}^{I_n} \mathcal{X}_{i_1, \ldots, i_N} \mathbf{U}_{j, i_n}.$$

Tucker decomposition [41] factorizes an N-way tensor into a core tensor multiplied by factor matrices along each mode. For a three-way tensor $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$, its Tucker decomposition is

$$\mathcal{X} = \mathcal{C} \times_1 \mathbf{A} \times_2 \mathbf{B} \times_3 \mathbf{D}$$

where $\mathcal{C} \in \mathbb{R}^{P \times Q \times R}$ is the core tensor and $\mathbf{A} \in \mathbb{R}^{I \times P}$, $\mathbf{B} \in \mathbb{R}^{J \times Q}$, and $\mathbf{D} \in \mathbb{R}^{K \times R}$ are the factor matrices along the first, second, and third modes, respectively. The core tensor captures interactions among the factors, and the factor matrices contain the basis vectors for each mode. The ranks (P, Q, R) control the compactness of the representation.

III. Method

A. Challenges and Motivations

While guided denoising offers a promising pathway for HSI restoration, developing a unified framework that can

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

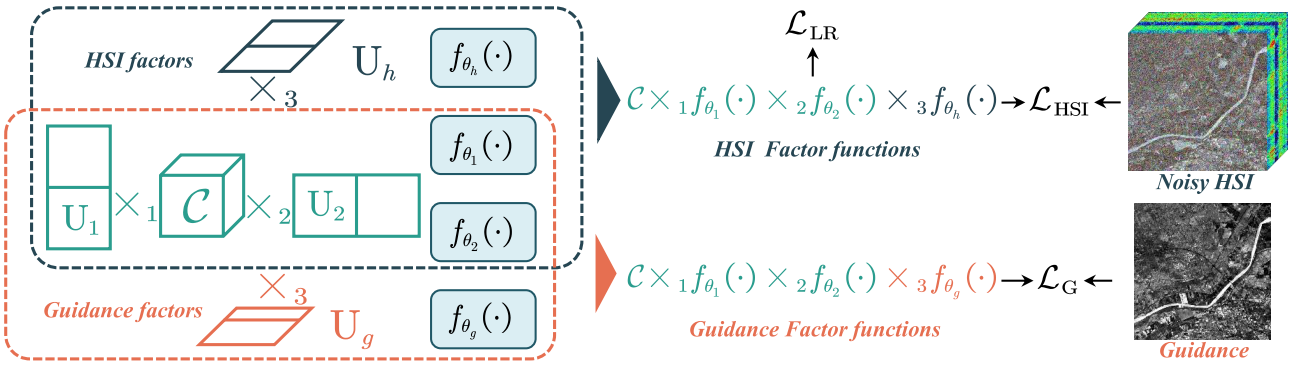


Fig. 2. Illustration of the proposed CCTD framework for HSI denoising.

incorporate diverse guidance modalities (e.g., PAN, MSI, and SAR) presents two fundamental challenges.

1) Texture Overtransferring: The different imaging mechanisms between HSIs and guidance modalities introduce a critical risk of texture overtransfer. Guidance modalities while sharing structural similarities, often contain unique textures that are absent in the target HSI. Directly imposing spatial priors from the guidance modality can inadvertently transfer these undesired structures, thereby corrupting the fine-grained spatial details that are intrinsic and unique to the HSI. A principled framework must therefore disentangle shared spatial structures from modality-specific characteristics.

2) Structural Heterogeneity: Guidance modalities differ fundamentally in their channel dimensions and imaging mechanisms. PAN is single-channel broadband, MSI provides several discrete spectral bands, and SAR encodes backscatter information. This heterogeneity creates a structural mismatch that precludes the direct application of conventional tensor operations or feature-space alignment techniques. Any unified framework must deal with this variability without forcing unnatural dimensionality reductions that discard valuable information.

Motivated by these challenges, a Tucker decomposition framework is proposed to model the multimodal data. The core insight is that Tucker decomposition naturally factorizes a tensor into a core tensor and factor matrices, where the factor matrices along the spatial modes capture spatial structure, and those along the spectral mode capture spectral signatures. By sharing the two spatial factor matrices across the HSI and guidance modalities, the framework achieves implicit spatial alignment while preserving modality-specific spectral characteristics through distinct spectral factors. This design simultaneously addresses all challenges: 1) the shared spatial factors enforce structural coherence while preventing texture overtransfer, as each modality retains its own spectral representation; and 2) the framework naturally accommodates arbitrary channel dimensions by projecting each modality into a shared spatial subspace. Sections III-B–III-D provide the formulation of this unified framework.

B. Coupled Tucker Decomposition

The proposed framework builds upon the classical tensor decomposition model—Tucker decomposition. For the restored HSI $\mathcal{H} \in \mathbb{R}^{H\times W\times B}$, where $H \times W$ denotes the spatial dimensions and B the number of spectral bands, Tucker decomposition factorizes it as

$$\hat{\mathcal{H}} = \mathcal{C} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \mathbf{U}_h \tag{1}$$

where $\times_n$ denotes the mode-n tensor-matrix product. Here, $\mathcal{C} \in \mathbb{R}^{h\times w\times b}$ is the core tensor that captures the multiway interactions among the decomposed dimensions; $\mathbf{U}_1 \in \mathbb{R}^{H\times h}$ and $\mathbf{U}_2 \in \mathbb{R}^{W\times w}$ are the spatial factor matrices encoding basis patterns along the height and width dimensions, respectively; and $\mathbf{U}_h \in \mathbb{R}^{B\times b}$ is the spectral factor matrix encoding the spectral basis. The dimensions (h, w, b) control the compactness of the representation.

This factorization can be intuitively understood as tensor dictionary learning. In particular, the factor matrices $\mathbf{U}_1, \mathbf{U}_2, \mathbf{U}_h$ constitute a separable, multilinear dictionary spanning the 3-D HSI space: each column of $\mathbf{U}_1$ (respectively, $\mathbf{U}_2$, $\mathbf{U}_h$) represents an atom capturing a fundamental pattern along the height (respectively width, spectral) dimension. The core tensor $\mathcal{C}$ serves as the coefficient tensor in a low-dimensional latent space: each entry c_{ijk} weights the interaction among the ith atom from $\mathbf{U}_1$, the jth atom from $\mathbf{U}_2$, and the kth atom from $\mathbf{U}_h$. Thus, the HSI is reconstructed as a multilinear combination of these dictionary atoms, with the core tensor governing the combination coefficients.

For a coregistered guidance modality $\hat{\mathcal{G}} \in \mathbb{R}^{H\times W\times B'}$ (e.g., MSI or SAR), a coupled decomposition is proposed that shares key components with the HSI representation

$$\hat{\mathcal{G}} = \mathcal{C} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \mathbf{U}_g \tag{2}$$

where $\mathbf{U}_g \in \mathbb{R}^{B'\times b'}$ is the guidance-specific channel factor matrix and b' denotes the rank along the channel mode for the guidance modality. Critically, the spatial factor matrices $\mathbf{U}_1, \mathbf{U}_2$ and the core tensor $\mathcal{C}$ are shared between the two decompositions. This sharing mechanism is the foundation of the unified guided denoising framework. Its design is based on three key ideas that come from how different types of remote sensing images work in practice.

1) *Shared Spatial Dictionaries:* The HSI and guidance modalities capture the same geographical scene, and thus share identical spatial structures, such as terrain contours, object boundaries, and textural patterns. The spatial factor matrices $\mathbf{U}_1$ and $\mathbf{U}_2$ serve precisely as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

the dictionaries that encode these spatial basis patterns. Sharing them across modalities enforces that both images are represented in the same spatial subspace, providing an implicit mechanism for spatial alignment and structural coherence. This design directly addresses the texture overtransfer risk (Challenge 1): by constraining the spatial representation to a shared subspace, such a constraint ensures that only the spatial structures present in both modalities are preserved, while modality-specific spatial artifacts are naturally suppressed.

2) *Shared Latent Representation:* The core tensor $\mathcal{C}$ acts as the joint latent representation that couples the two modalities. From the dictionary learning perspective, $\mathcal{C}$ encodes the coefficients that weight the interaction of spatial and spectral atoms. Sharing $\mathcal{C}$ imposes that the two modalities share the same underlying "semantic" structure in the joint spatial–spectral latent space. This facilitates cross-modal information fusion: the guidance modality contributes to estimating a clean, noise-free core tensor, which in turn reconstructs a denoised HSI via (1). This coupling is the mechanism through which the external prior from the guidance modality is effectively transferred to the HSI restoration task.
3) *Modality-Specific Spectral Dictionaries:* In contrast to the spatial components, the spectral factor matrices $\mathbf{U}_h$ (for HSI) and $\mathbf{U}_g$ (for guidance) must remain modality-specific. This is necessitated by fundamental differences in the imaging physics: HSIs possess high spectral resolution with B narrow, contiguous bands; MSIs have fewer, broader, and often nonoverlapping bands; SAR images encode backscatter intensity with entirely different physical semantics. Sharing spectral dictionaries would force an unnatural alignment of these disparate physical quantities, leading to the negative transfer of spectral information and degraded performance. By keeping $\mathbf{U}_h$ and $\mathbf{U}_g$ independent, the framework respects the intrinsic spectral uniqueness of each modality while leveraging the shared spatial structure for guided denoising.

The joint decomposition formulated in (1) and (2) is termed coupled Tucker decomposition (CTD). The CTD framework elegantly addresses all the aforementioned challenges. The shared spatial factors prevent texture overtransfer while ensuring spatial coherence, and the formulation naturally accommodates arbitrary channel dimensions (B and B') through modality-specific spectral factors.

C. Continuous CTD

The CTD introduced above establishes a shared latent space that leverages external priors from guidance modalities. However, relying solely on external information is insufficient for high-quality HSI restoration. The target HSI itself possesses rich internal priors that must be exploited. Among these, local spatial smoothness (i.e., the tendency of neighboring pixels to exhibit similar intensities) is particularly important for suppressing noise while preserving edges. Traditional approaches enforce this prior by augmenting the objective with a TV regularizer. However, TV regularization introduces a critical practical challenge: its associated hyperparameter, which balances the tradeoff between data fidelity and smoothness, is very difficult to tune. This difficulty comes from the hyperparameter's strong coupling with the magnitudes of other loss terms, necessitating extensive efforts.

To address this limitation, this article proposes to model the smoothness prior directly into the representation itself, rather than enforcing it through an explicit regularizer. The key insight is to parameterize the factor matrices as continuous functions of spatial coordinates using INRs. This approach, inspired by recent advances in coordinate-based MLPs [29], [30], offers two fundamental advantages: 1) it naturally encodes spatial smoothness through the network's architectural inductive bias and 2) it decouples smoothness control from the loss function, replacing a difficult-to-tune hyperparameter with an interpretable and easily adjustable frequency parameter.

Concretely, each column of the factor matrices is treated as a sample from an underlying continuous function. For $\mathbf{U}_1 \in \mathbb{R}^{H\times h}$ (where rows correspond to height coordinates and columns to atoms), a mapping is defined to associate each height coordinate to the corresponding row vector in $\mathbf{U}_1$. This is achieved by learning a function $f_{\theta_1} : \mathbb{R} \to \mathbb{R}^h$ parameterized by a multilayer perceptron (MLP)

$$\mathbf{U}_1 = f_{\theta_1}(\mathbf{x}), \quad \text{with } \mathbf{x} = [1, 2, \ldots, H]^T \tag{3}$$

where $\mathbf{x}$ is the vector of normalized height coordinates and θ_1 denotes the learnable parameters of the MLP. The function f_{θ_1} is evaluated at each coordinate to produce the corresponding row of $\mathbf{U}_1$. Analogously, f_{θ_2} is defined for the width dimension, and f_{θ_h}, f_{θ_g} for the spectral dimensions of the HSI and guidance modality, respectively.

Significantly, the choice of activation function in the MLP governs the smoothness of the learned representation. Following [29], this framework adopts the sine activation:

$$\sigma(z) = \sin(\omega z) \tag{4}$$

where ω is a frequency hyperparameter that controls the bandwidth of the representation. A larger ω enables the network to represent higher frequency details, while a smaller ω imposes a smoother, low-frequency bias. This property provides a direct and interpretable mechanism for enforcing local smoothness: by setting an appropriate ω, the spatial factor matrices can be implicitly regularized to be smooth functions of the coordinates. Unlike the TV hyperparameter, ω is decoupled from the loss function, since it is a property of the network architecture itself. This decoupling substantially simplifies hyperparameter selection; in practice, setting $\omega = 3$ is found to yield an effective balance between preserving fine spatial details and ensuring sufficient smoothness for natural images, consistent with observations in prior work [29].

Substituting these continuous factor matrices into the CTD formulation yields the final framework, which is termed CCTD

$$\begin{aligned}\hat{\mathcal{H}} &= \mathcal{C} \times_1 f_{\theta_1}(\mathbf{x}) \times_2 f_{\theta_2}(\mathbf{y}) \times_3 f_{\theta_h}(\mathbf{z}_h)\\ \hat{\mathcal{G}} &= \mathcal{C} \times_1 f_{\theta_1}(\mathbf{x}) \times_2 f_{\theta_2}(\mathbf{y}) \times_3 f_{\theta_g}(\mathbf{z}_g)\end{aligned} \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

where $\mathbf{y} = [1, 2, \ldots, W]^T$ and $\mathbf{z}_h = [1, 2, \ldots, B]^T, \mathbf{z}_g = [1, 2, \ldots, B']^T$ are the coordinates along the width and spectral dimensions, respectively. The continuous reconstruction $\hat{\mathcal{H}}$ is mapped back to the pixel domain, yielding the final denoised HSI output $\mathcal{X} \in \mathbb{R}^{H \times W \times B}$. CCTD naturally integrates external priors (via shared core tensor and spatial factors) with internal priors (via the implicit smoothness of the continuous spatial representations), providing a principled and unified solution for guided HSI denoising. The learnable parameters now comprise the core tensor $\mathcal{C}$ and the MLP weights $\theta_1, \theta_2, \theta_h, \theta_g$, all of which are optimized jointly to minimize a reconstruction loss, as detailed in Sections III-B–III-D.

D. Objective Function and Optimization

With the CCTD formulation established, the objective function is defined for guided HSI denoising. Loss function comprises three terms, each targeting a specific aspect of the restoration problem: fidelity to the observed noisy HSI, fidelity to the clean guidance modality, and a low-rank prior to exploit spectral correlations.

1) HSI Fidelity Term: The primary objective is to ensure that the reconstructed HSI $\hat{\mathcal{H}}$ closely approximates the observed noisy HSI $\mathcal{N}$. This is formulated as

$$\mathcal{L}_{\text{HSI}} = \left\| \hat{\mathcal{H}} - \mathcal{N} \right\|_p^p \tag{6}$$

where $\|\cdot\|_p$ denotes the ℓ_p norm. The choice of p is dictated by the noise characteristics: for additive Gaussian noise, the ℓ_2 norm (corresponding to the negative log-likelihood under a Gaussian noise model) is optimal; for mixed noise containing outliers or heavy-tailed components, the ℓ_1 norm provides greater robustness. In the experiments, ℓ_2 is adopted for synthetically corrupted datasets and ℓ_1 for real-world noisy data where the noise distribution is unknown and potentially non-Gaussian.

2) Guidance Fidelity Term: The guidance modality $\mathcal{G}$ is assumed to be clean or significantly less noisy than the HSI. To ensure structural consistency, the reconstructed guidance modality $\hat{\mathcal{G}}$ is constrained to match the observed guidance modality, which serves as a robust external prior. This consistency is enforced by minimizing the squared Frobenius norm

$$\mathcal{L}_{\text{G}} = \left\| \hat{\mathcal{G}} - \mathcal{G} \right\|_F^2. \tag{7}$$

This term ensures that the shared spatial factors $\mathbf{U}_1, \mathbf{U}_2$ and the core tensor $\mathcal{C}$ are informed by the high-quality structural information from the guidance modality, thereby guiding the HSI reconstruction toward spatially coherent solutions.

3) Low-Rank Regularization: Tucker decomposition inherently imposes a low-rank prior through the choice of core tensor dimensions (h, w, b). In principle, setting these dimensions sufficiently small enforces a low-rank approximation of the HSI. However, selecting appropriate ranks is notoriously difficult: overly aggressive rank reduction can suppress fine spectral details and degrade representation capacity, while insufficient reduction fails to remove noise effectively. Manual tuning of three coupled hyperparameters is impractical and dataset-dependent.

To avoid this limitation, an explicit low-rank regularizer is introduced that operates directly on the reconstructed HSI, allowing the model to separate rank selection from the optimization process. In particular, the nuclear norm is applied to the mode-3 (spectral) unfolding of $\hat{\mathcal{H}}$:

$$\mathcal{L}_{\text{LR}} = \left\| \hat{\mathcal{H}} \right\|_* = \sum_{k=1}^{\min(B, HW)} \sigma_k \left(\hat{\mathbf{H}}_{(3)} \right) \tag{8}$$

where $\hat{\mathbf{H}}_{(3)} \in \mathbb{R}^{B \times HW}$ is the mode-3 unfolding of $\hat{\mathcal{H}}$, and σ_k denotes its singular values. Minimizing the nuclear norm encourages the reconstructed HSI to exhibit low-rank structure along the spectral dimension, which is a well-established prior for HSIs due to high spectral correlation. Significantly, this formulation eliminates the need to manually tune the Tucker ranks (h, w, b); they can be set sufficiently large to preserve representation capacity while relying on the nuclear norm to enforce a spectral low-rank prior. The ranks become architectural parameters rather than critical hyperparameters, substantially simplifying model selection.

In summary, combining the three terms, the complete loss function is

$$\mathcal{L} = \mathcal{L}_{\text{HSI}} + \lambda_1 \mathcal{L}_{\text{G}} + \lambda_2 \mathcal{L}_{\text{LR}} \tag{9}$$

where λ_1 and λ_2 are positive hyperparameters balancing the contributions of the guidance prior and the low-rank regularization, respectively.

As for optimization, all learnable parameters (the core tensor $\mathcal{C}$ and the MLP weights $\theta_1, \theta_2, \theta_h, \theta_g$) are optimized jointly by minimizing $\mathcal{L}$ using stochastic gradient descent or its variants (e.g., Adam). The gradients are computed via automatic differentiation, and the Tucker reconstructions $\hat{\mathcal{H}}$ and $\hat{\mathcal{G}}$ are formed on-the-fly during each forward pass. The workflow is summarized in Algorithm 1.

Algorithm 1 CCTD for Guided HSI Denoising

Input: Noisy HSI $\mathcal{N} \in \mathbb{R}^{H \times W \times B}$, Guidance modality $\mathcal{G} \in \mathbb{R}^{H \times W \times B'}$ (PAN/MSI/SAR), Tucker ranks (r_1, r_2, r_3), tradeoff parameters λ_1, λ_2.
Output: Denoised HSI $\mathcal{X}$.
Step 1: Continuous Factor Parameterization
· Initialize $\mathcal{C}, \theta_1, \theta_2, \theta_h, \theta_g$;
· Parameterize spatial factors $\mathbf{U}_1 \in \mathbb{R}^{H \times r_1}$, $\mathbf{U}_2 \in \mathbb{R}^{W \times r_2}$ and spectral factors $\mathbf{U}_h \in \mathbb{R}^{B \times r_3}$, $\mathbf{U}_g \in \mathbb{R}^{B' \times r_3}$ via INRs: $\mathbf{U}_1 = f_{\theta_1}(\mathbf{x}), \mathbf{U}_2 = f_{\theta_2}(\mathbf{y}), \mathbf{U}_h = f_{\theta_h}(\mathbf{z}_h), \mathbf{U}_g = f_{\theta_g}(\mathbf{z}_g)$;
Step 2: Coupled Continuous Modeling
· Formulate HSI representation: $\hat{\mathcal{H}} = \mathcal{C} \times_1 f_{\theta_1}(\mathbf{x}) \times_2 f_{\theta_2}(\mathbf{y}) \times_3 f_{\theta_h}(\mathbf{z}_h)$;
· Formulate Guidance representation: $\hat{\mathcal{G}} = \mathcal{C} \times_1 f_{\theta_1}(\mathbf{x}) \times_2 f_{\theta_2}(\mathbf{y}) \times_3 f_{\theta_g}(\mathbf{z}_g)$;
Step 3: Joint Optimization
· Objective Function: $\mathcal{L} = \|\hat{\mathcal{H}} - \mathcal{N}\|_P^p + \lambda_1 \|\hat{\mathcal{G}} - \mathcal{G}\|_F^2 + \lambda_2 \sum_{k=1}^{\min(B, HW)} \sigma_k(\hat{\mathbf{H}}_{(3)})$;
· Update $\{\mathcal{C}, \theta_1, \theta_2, \theta_h, \theta_g\}$ to minimize $\mathcal{L}$ by Adam;
Step 4: HSI Reconstruction
· Evaluate the continuous spatial factors at all pixel coordinates;
· Reconstruct final denoised HSI: $\mathcal{X} = \mathcal{C} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \times_3 \mathbf{U}_h$;
Return: $\mathcal{X}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

E. Computational Complexity

In the CCTD framework, the continuous representation is decomposed into multiple factor functions along spatial and spectral modes. In particular, instead of feeding dense 3-D coordinates, CCTD only requires four low-dimensional coordinate vectors corresponding to the height, width, and spectral dimensions of the HSI and guidance modality, i.e., of sizes $H \times 1$, $W \times 1$, $B \times 1$, and $B' \times 1$.

As a result, the computational complexity of CCTD in each forward pass is

$$\mathcal{O}\left(m\hat{r}d\left(H+W+B+B'\right)+\hat{r}HW\left(B+B'\right)\right) \tag{10}$$

where $\hat{r} = \max\{r_1, r_2, r_3\}$, m denotes the number of hidden units and d denotes the network depth. The first term corresponds to the evaluation of the factor functions, while the second term arises from the Tucker reconstruction with the shared core tensor and spatial functions.

IV. Experiments

A. Datasets

The proposed CCTD is evaluated on two multimodal remote sensing datasets covering different guidance modalities.

1) *C2Seg-BW (Beijing, Wuhan) [42], [43]:* This dataset contains HSI, MSI, and SAR images acquired by the Gaofen-5 (HSI), Gaofen-6 (MSI), and Gaofen-3 (dual-Pol SAR with HH and HV channels) satellite missions over Beijing and Wuhan, China. The images are captured on dates as close as possible to minimize seasonal variations. For experiments, 256×256 pixel patches were extracted from each city.
2) *Florence:* This dataset is acquired by the PRISMA satellite mission, which carries both an HSI sensor and a PAN sensor. The scene covers a rural area near Florence, Italy, with 256×256 pixels.

All images are normalized to $[0, 1]$ before processing. For datasets with multiple guidance modalities (Beijing and Wuhan), each guidance type (MSI and SAR) is treated independently, i.e., separate experiments are conducted for MSI-guided and SAR-guided denoising to evaluate the individual contribution of each guidance type. All the guidance modalities are resized to match the spatial dimensions of the HSI.

Due to the different modalities, when the CCTD algorithm adopts modality X, we denote it as CCTD(X), where $X = \text{PAN}, \text{MSI}, \text{SAR}$. For example, CCTD(MSI) indicates that MSI is utilized as the guidance modality. So does PWRCTV and DTucker.

B. Experimental Setup

1) Compared Methods and Metrics: The compared methods for HSI denoising include NGmeet [44], BALMF [31], LRTV [12], LMHTV [45], LTHTV [45], TCTV [37], RCTV [39], CTV [38], HIRDiff [46], LRTFR [47], CRNL [48], PWRCTV [18], and DTucker [40]. Among these denoising methods, CCTD, PWRCTV, and DTucker are multimodal approaches, while the others are single-modal approaches. Four widely used metrics are employed for HSI quality assessment: peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) [49], erreur relative globale adimensionnelle de synthése (ERGAS) [50], and spectral angle mapper (SAM) [51]. Higher PSNR/SSIM and lower ERGAS/SAM indicate better performance.

2) Noise Simulation: To evaluate the robustness of the proposed method under complex degradations, six noise scenarios are simulated with increasing severity. The noise is applied exclusively to the HSI, while the guidance modality remains clean. The specific configurations are as follows.

Case 1: Only i.i.d. Gaussian noise with standard deviation 10/255.

Case 2: Mixed noise including non-i.i.d. Gaussian noise (band-dependent σ uniformly sampled from $[5/255, 50/255]$), impulse noise (salt-and-pepper with probability 0.2), deadline noise (vertical/horizontal missing lines with probability 0.2), and stripe noise (vertical/horizontal stripes with probability 0.2). Each noise type corrupts a randomly selected subset of bands, covering 2/3 of the bands for Gaussian and 1/3 for each of the other types.

Case 3: Same as Case 2, but with increased probability to 0.3 for impulse/deadline/stripe.

Case 4: Probabilities increased to 0.4.

Case 5: Probabilities increased to 0.5.

Case 6: Probabilities are randomly sampled from $[0.05, 0.5]$ for impulse/deadline/stripe, while Gaussian noise remains non-i.i.d. with $\sigma \in [5/255, 50/255]$.

These cases simulate realistic, complex noise scenarios encountered in hyperspectral imaging, including band-dependent Gaussian, impulse, stripes, and deadlines.

3) Parameter Configuration and Training: The CCTD framework is implemented using PyTorch and executed on an NVIDIA GeForce RTX 4090 GPU. The INR networks (f_{θ_1}, f_{θ_2}, f_{θ_h}, and f_{θ_g}) are parameterized by MLPs consisting of three hidden layers with a width of 300 units, utilizing sine activation functions ($\omega = 3$). The core tensor $\mathcal{C}$ is initialized randomly and optimized jointly. Model training is conducted for 5000 iterations using the Adam optimizer with a learning rate of 10^{-4}. For the HSI reconstruction loss $\mathcal{L}_{\text{HSI}}$, ℓ_2 is employed for Case 1 while ℓ_1 is employed for Cases 2–6. The loss weights are set as follows: $(\lambda_1, \lambda_2) = (0.1, 0.04)$ for Case 1 and $(\lambda_1, \lambda_2) = (200, 90)$ for other cases.

Regarding the baseline methods, PWRCTV was originally designed for PAN-guided HSI denoising. For multichannel modalities, it is adapted to MSI and SAR by averaging all available channels to generate a single-channel guidance modality. All compared methods are implemented using publicly available codes and tuned for optimal performance on each dataset.

C. Quantitative Results

Quantitative results on the Beijing, Wuhan, and Florence datasets under the six noise cases are reported. Tables I–III summarize the PSNR, SSIM, ERGAS, and SAM. The key observations are:

CCTD generally outperforms all competing methods across all datasets and noise levels. For example, in Beijing under

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

TABLE I
METRICS FOR HSI DENOISING ON BEIJING DATASET. THE BEST AND THE SECOND BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Algorithm	Case 1				Case 2				Case 3			
	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓
NGMeet	**41.88**	**0.9950**	**0.53**	**1.31**	24.97	0.7949	6.75	15.81	23.53	0.7583	8.20	18.86
BALMF	36.35	0.9793	1.04	3.08	31.82	0.9105	5.14	13.25	29.87	0.8748	6.45	16.78
LRTV	33.77	0.9689	1.36	2.51	28.37	0.8777	4.84	11.61	26.34	0.8314	6.55	14.90
LMHTV	38.10	0.9869	0.86	2.36	31.62	0.9089	5.16	13.19	29.92	0.8925	6.00	15.22
LTHTV	38.26	0.9879	0.82	2.17	30.85	0.9062	4.87	12.15	29.04	0.8806	6.06	14.95
TCTV	36.00	0.9812	1.05	3.23	27.85	0.8296	6.07	13.62	26.17	0.7791	7.65	17.27
RCTV	39.56	0.9916	0.69	1.66	32.11	0.9545	2.52	6.01	29.77	0.9161	4.41	9.86
CTV	38.89	0.9901	0.76	1.89	35.27	0.9729	1.62	4.33	30.84	0.9087	3.91	9.33
LRTFR	34.35	0.9796	1.27	2.26	31.87	0.9568	2.20	5.55	31.06	0.9295	3.41	8.63
CRNL	32.93	0.9743	1.47	1.79	26.77	0.9056	3.83	8.08	25.12	0.8610	5.26	11.27
HIRDiff	35.79	0.9756	1.07	2.10	23.87	0.7991	4.66	9.85	19.31	0.6139	7.43	13.70
DTucker(MSI)	39.24	0.9906	0.73	1.76	29.89	0.9218	4.08	10.05	27.56	0.8606	5.97	14.23
DTucker(SAR)	39.06	0.9905	0.74	1.79	29.91	0.9217	4.08	10.03	27.59	0.8625	5.89	14.11
PWRCTV(MSI)	39.13	0.9908	0.73	1.67	31.82	0.9338	3.71	8.63	29.54	0.8836	5.39	12.33
PWRCTV(SAR)	39.13	0.9904	0.73	1.74	31.89	0.9293	3.92	9.35	29.70	0.8809	5.58	12.84
CCTD(MSI)	39.42	0.9909	0.71	2.01	**36.43**	**0.9857**	**1.13**	**3.33**	**35.46**	**0.9828**	**1.27**	**3.64**
CCTD(SAR)	39.34	0.9907	0.72	2.02	36.18	0.9848	1.16	3.39	35.26	0.9819	1.29	3.69

Algorithm	Case 4				Case 5				Case 6			
	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓
NGMeet	22.80	0.7032	9.42	22.12	22.57	0.6985	9.90	23.03	24.99	0.7796	9.00	21.39
BALMF	25.72	0.7521	11.49	27.36	21.90	0.5940	16.08	38.25	29.89	0.8771	7.50	19.02
LRTV	23.70	0.7243	10.45	22.87	22.95	0.6740	13.10	28.89	26.95	0.8403	7.01	15.79
LMHTV	27.08	0.8173	8.00	19.02	23.78	0.7033	12.22	29.35	29.84	0.8807	7.15	18.32
LTHTV	26.09	0.8261	8.14	20.21	23.44	0.7057	11.80	27.96	28.13	0.8620	6.98	17.87
TCTV	24.81	0.7378	9.04	20.45	23.71	0.7020	10.21	23.08	26.56	0.7840	7.77	17.55
RCTV	26.84	0.8191	8.18	17.88	24.92	0.7262	11.88	26.08	29.90	0.8988	5.51	12.44
CTV	28.54	0.8310	6.84	15.32	27.03	0.7646	9.51	20.56	32.21	0.8931	4.90	11.40
LRTFR	29.32	0.9076	4.55	11.17	28.25	0.8814	5.55	13.58	30.44	0.9321	3.83	9.94
CRNL	23.76	0.8141	6.69	14.52	22.64	0.7569	8.18	17.70	25.86	0.8537	5.70	12.80
HIRDiff	22.46	0.7701	6.82	14.75	22.50	0.7250	8.47	19.18	23.54	0.7870	6.02	13.71
DTucker(MSI)	25.43	0.7724	8.71	19.63	23.73	0.6987	11.52	26.31	28.03	0.8707	5.96	14.62
DTucker(SAR)	25.40	0.7751	8.56	19.38	23.68	0.7005	11.43	26.16	28.08	0.8775	5.83	14.46
PWRCTV(MSI)	26.88	0.8139	8.12	17.89	25.31	0.7286	11.51	25.28	29.56	0.8737	6.04	13.64
PWRCTV(SAR)	26.91	0.8075	8.09	17.66	25.04	0.7219	11.45	25.14	29.75	0.8753	5.99	13.86
CCTD(MSI)	**34.32**	**0.9791**	**1.47**	**4.09**	**32.29**	0.9682	2.09	5.50	**35.62**	**0.9836**	1.32	3.88
CCTD(SAR)	34.18	0.9786	1.48	4.10	32.26	**0.9694**	**2.06**	**5.39**	35.44	0.9835	**1.32**	**3.85**

Case 5 (severe mixed noise), CCTD improves PSNR by over 3 dB compared to the best single-HSI method and by 1.5 dB compared to PWRCTV. As shown in noise cases 2–5, while the performance of single-HSI methods degrades rapidly as noise increases, CCTD maintains high fidelity due to the clean guidance prior and the continuous representation that enforces spatial smoothness. Compared with the multimodal method, CCTD works equally well with MSI and SAR guidance on Beijing/Wuhan, and with PAN guidance on Florence. PWRCTV, when adapted to MSI/SAR, shows only marginal improvement over single-HSI methods, indicating the importance of a modality-agnostic design.

Fig. 3 shows visual comparisons on the Wuhan dataset under Case 4 noise, CCTD utilized MSI and SAR as the guidance modalities, respectively. The MSI is visualized with red, green, and blue channels. As for SAR, there are only HH and HV channels, and for visualization, we regard the average of HH and HV channels as the third channel. The noisy HSI exhibits severe stripes and impulse corruption. Single-HSI methods fail to remove all artifacts or tend to oversmooth textures. PWRCTV (with SAR guidance) preserves edges better but still leaves residual stripes. On the contrary, CCTD produces a clean reconstruction that closely resembles the ground truth, with sharp edges and fine details well preserved. The advantage is also revealed by spectral profiles in Fig. 5, where blue, red, and gray curves represent the ground truth, denoised, and noisy spectral vectors, respectively. CCTD more accurately recovers the original spectral signatures, while other methods display significant fluctuations, implying better spectral reconstruction. Moreover, Fig. 6 presents visual comparisons on the Florence dataset under Case 3 noise. Using the PAN image as the guidance modality, CCTD achieves a remarkable denoising effect. This observation demonstrates that CCTD remains highly effective even when guided by single-channel modalities.

D. Effect of Single-Channel Guidance

To further ensure a fair comparison with PWRCTV, which is designed for single-channel guidance, the same generation strategy is adopted to compress the multichannel guidance modalities into single-channel modalities for CCTD. The corresponding CCTD variant is denoted as CCTD-SCG (**S**ingle-**C**hannel **G**uidance). As shown in Table IV,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

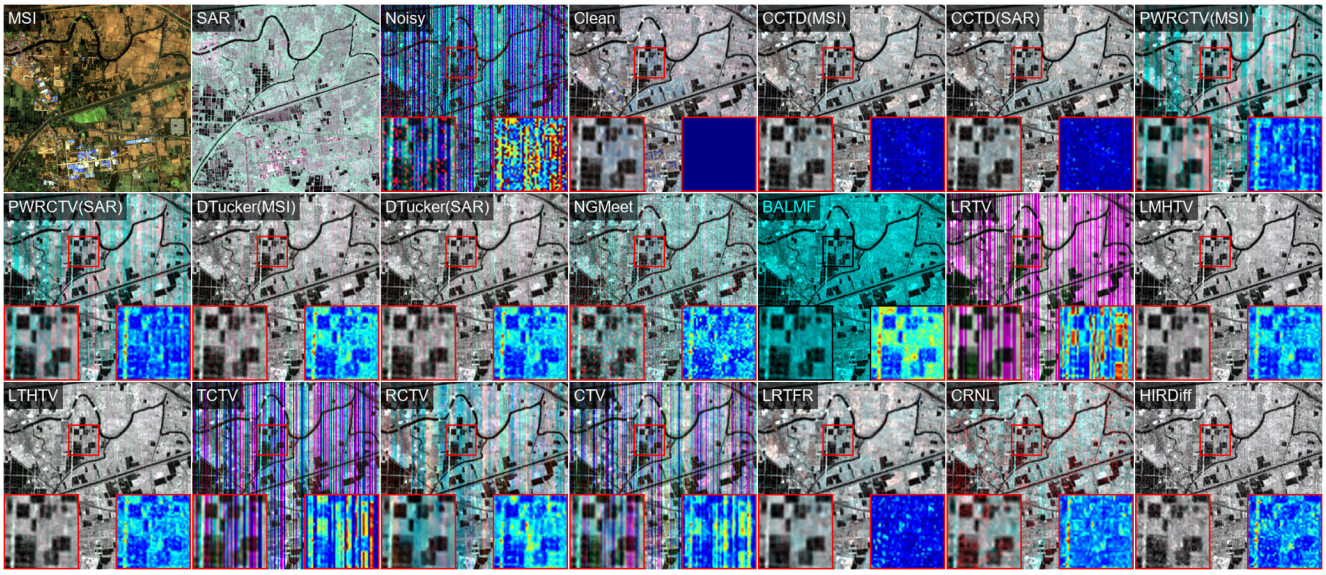


Fig. 3. Denoising visual comparisons on Wuhan dataset under noise case 4 (Bands: 17-11-8).

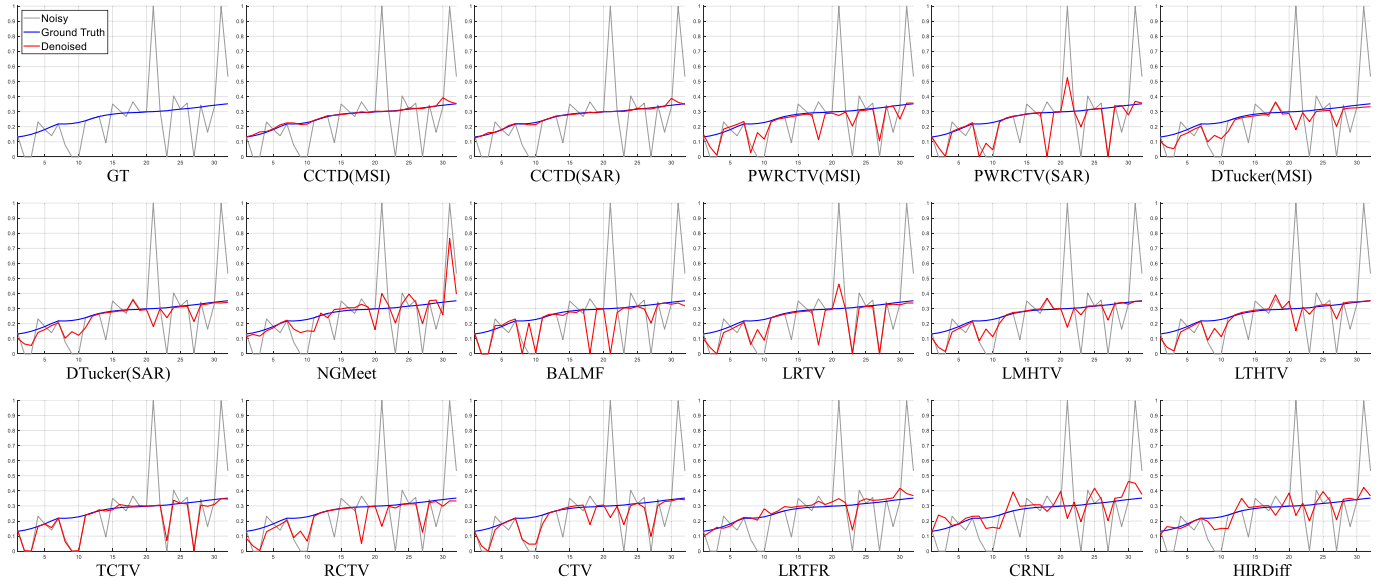


Fig. 4. PSNR robustness comparison of CCTD, PWRCTV, and DTucker on Beijing dataset under noise Case 1, with Gaussian noise level σ in the guidance modality varying from 0 to 0.1.

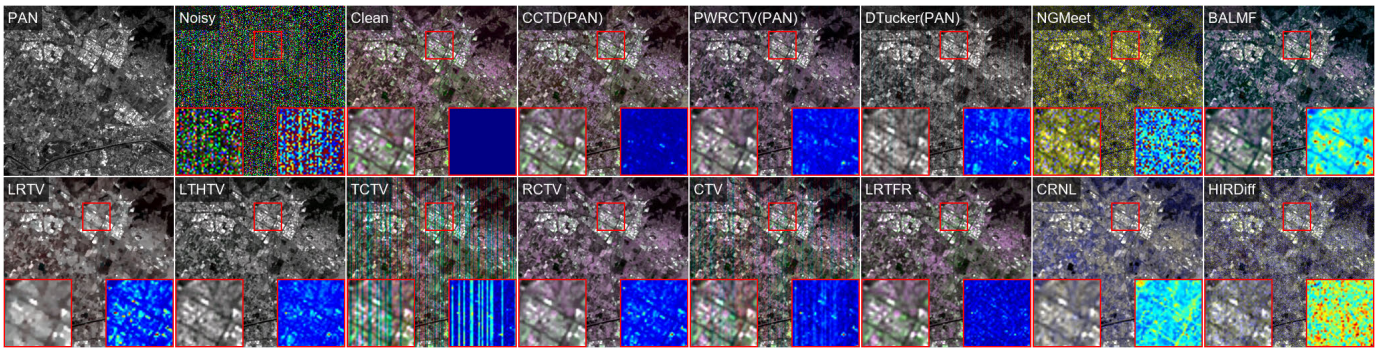


Fig. 5. Spectral signature curves of pixel (50, 50) reconstructed by all the compared methods on the Wuhan dataset under noise Case 4.

CCTD-SCG suffers a slight performance drop after channel compression, but still outperforms PWRCTV in most cases. This multiscale comparison further demonstrates the superiority of CCTD over previous guided denoising methods.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

TABLE II

METRICS FOR HSI DENOISING ON WUHAN DATASET. THE BEST AND THE SECOND BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Algorithm	Case 1				Case 2				Case 3			
	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓
NGMeet	**41.27**	**0.9863**	**1.00**	**1.55**	24.97	0.7006	11.06	17.66	23.29	0.6403	11.98	20.90
BALMF	36.21	0.9525	1.97	3.49	31.00	0.8693	6.20	14.05	28.35	0.7912	9.42	18.36
LRTV	34.71	0.9414	1.87	2.38	30.28	0.8637	4.39	9.86	26.50	0.7427	7.67	15.73
LMHTV	38.21	0.9710	1.49	2.61	30.41	0.8402	6.67	15.26	28.97	0.8174	7.63	17.15
LTHTV	38.44	0.9724	1.43	2.49	29.38	0.8224	6.65	14.90	27.99	0.7957	7.94	17.55
TCTV	34.96	0.9449	2.28	4.54	28.74	0.7831	6.50	13.34	27.45	0.7342	7.92	16.39
RCTV	38.89	0.9773	1.32	1.89	32.46	0.9291	3.05	5.84	30.73	0.8910	4.79	8.74
CTV	38.21	0.9742	1.54	2.28	34.24	0.9415	2.75	5.40	30.82	0.8697	4.88	9.50
LRTFR	34.60	0.9580	2.00	2.57	32.44	0.9108	3.29	6.93	31.22	0.8930	4.33	8.84
CRNL	34.37	0.9532	1.93	1.97	27.70	0.8293	5.07	8.41	26.02	0.7774	6.61	11.48
HIRDiff	35.25	0.9423	1.79	2.11	26.55	0.7867	5.45	9.41	25.11	0.7513	6.45	11.54
DTucker(MSI)	38.35	0.9728	1.54	2.15	28.75	0.8302	5.73	11.81	27.25	0.8009	6.94	14.50
DTucker(SAR)	38.22	0.9727	1.55	2.13	28.81	0.8305	5.72	11.80	27.30	0.8021	6.90	14.45
PWRCTV(MSI)	38.83	0.9755	1.30	1.82	33.71	0.9318	3.15	6.43	30.58	0.8654	5.52	10.67
PWRCTV(SAR)	38.76	0.9753	1.33	1.98	32.08	0.9039	3.82	8.07	30.13	0.8525	5.79	11.52
CCTD(MSI)	39.25	0.9765	1.34	2.11	**36.36**	**0.9723**	**1.76**	**3.05**	**35.20**	**0.9648**	**2.00**	**3.48**
CCTD(SAR)	39.18	0.9762	1.35	2.13	36.07	0.9709	1.81	3.16	35.03	0.9639	2.03	3.56

Algorithm	Case 4				Case 5				Case 6			
	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓
NGMeet	22.35	0.6032	11.35	23.75	21.71	0.5781	12.64	27.16	24.73	0.6907	14.19	24.76
BALMF	25.44	0.6697	13.26	28.07	22.27	0.5316	17.23	38.40	27.52	0.7682	11.67	23.54
LRTV	24.78	0.6612	10.89	20.89	23.95	0.5980	13.32	26.02	26.94	0.7464	8.17	17.96
LMHTV	26.22	0.7500	9.76	20.70	24.01	0.6408	13.34	29.51	28.70	0.8056	10.44	21.63
LTHTV	25.59	0.7307	9.92	20.87	22.97	0.6082	13.32	28.54	27.31	0.7761	10.13	21.13
TCTV	26.39	0.6941	9.18	19.03	25.51	0.6607	10.28	21.27	28.01	0.7396	8.15	18.24
RCTV	27.26	0.7631	8.92	15.98	25.56	0.6783	11.96	22.82	29.58	0.8397	6.36	13.61
CTV	28.83	0.7952	7.46	14.39	27.46	0.7340	9.85	18.78	31.86	0.8510	5.78	13.02
LRTFR	30.24	0.8795	5.54	11.31	27.33	0.7971	7.96	16.05	31.18	0.8894	5.04	11.87
CRNL	24.92	0.7342	7.64	14.40	24.07	0.6914	8.47	16.98	26.48	0.7730	7.28	14.70
HIRDiff	23.68	0.6950	7.62	14.24	22.52	0.6460	8.69	16.93	25.19	0.7375	7.71	14.42
DTucker(MSI)	25.69	0.7413	8.85	18.69	23.62	0.6542	11.76	25.67	26.96	0.7745	8.88	17.36
DTucker(SAR)	25.72	0.7445	8.77	18.56	23.62	0.6588	11.65	25.38	27.00	0.7772	8.92	17.31
PWRCTV(MSI)	27.96	0.7793	8.79	16.46	25.84	0.6784	11.97	23.03	29.95	0.8359	6.59	14.53
PWRCTV(SAR)	27.62	0.7653	9.00	17.34	24.91	0.6564	12.37	24.91	30.20	0.8436	6.37	14.39
CCTD(MSI)	**33.93**	**0.9550**	**2.35**	**4.13**	**32.23**	**0.9397**	**3.05**	**5.48**	35.54	0.9686	**2.02**	3.78
CCTD(SAR)	33.82	0.9542	2.38	4.18	32.16	0.9390	3.07	5.51	**35.56**	**0.9688**	2.02	**3.76**

E. Ablation Studies

Ablation experiments are conducted to validate the rationality of CCTD. The analysis is divided into two parts, focusing on the loss function design and the modeling strategy, respectively. The corresponding results are presented in Tables V and VI on Beijing dataset under noise Case 6.

1) Loss Functions: The ablation results in Table V verify the effectiveness of the proposed loss function. Comparing experiments (c) and (e) shows that the guidance fidelity term $\mathcal{L}_{\mathrm{G}}$ significantly improves the denoising performance, demonstrating that the guidance modality provides reliable spatial constraints for restoring degraded HSI structures. The comparison between experiments (b) and (e) confirms the contribution of the low-rank regularization term $\mathcal{L}_{\mathrm{LR}}$, which effectively exploits the intrinsic low-rank redundancy of HSIs and reduces the dependence on manually selected Tucker ranks. Moreover, the performance gap between experiments (d) and (e) indicates that coupled optimization is essential for effective cross-modal information transfer. These results demonstrate that the proposed loss function jointly exploits reconstruction fidelity, guidance consistency, and low-rank prior, leading to more accurate and robust HSI denoising.

2) Different Low-Rank Regularization Terms: The nuclear norm is imposed on the mode-3 unfolding matrix to exploit the spectral low-rank property. To further analyze the choice of the low-rank regularization term, a tensor nuclear norm-based strategy is evaluated to jointly constrain the spatial and spectral dimensions. The results in Table VI show that this joint constraint leads to performance degradation, especially in terms of PSNR. This indicates that the spatial dimensions exhibit a weaker low-rank property than the spectral dimension. Therefore, applying nuclear norm constraints to the spatial dimensions may overconstrain the spatial representation and weaken the transfer of multimodal guidance information.

3) Spectral Coupling Strategy: To further investigate different multimodal guidance strategies, a spectral coupling strategy is evaluated by exploiting the flexibility of INR to share spectral factors across modalities. The results in Table VI show that this strategy causes performance degradation to varying degrees, especially in terms of SAM. This suggests

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

TABLE III

METRICS FOR HSI DENOISING ON FLORENCE DATASET. THE BEST AND THE SECOND BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Algorithm	Case 1				Case 2				Case 3			
	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓
NGMeet	**41.78**	**0.9835**	**1.25**	**1.81**	26.24	0.7035	12.70	16.35	24.04	0.6113	16.94	20.72
BALMF	37.95	0.9604	1.97	2.86	31.09	0.8381	10.16	12.19	29.36	0.7741	12.54	17.22
LRTV	34.84	0.9145	2.75	3.11	30.66	0.8162	7.21	9.19	27.64	0.7145	9.69	14.46
LMHTV	39.22	0.9723	1.71	2.44	31.32	0.8712	6.32	10.79	29.51	0.8177	8.99	13.77
LTHTV	39.16	0.9718	1.71	2.40	31.08	0.8607	6.24	10.34	29.42	0.8061	9.04	12.57
TCTV	35.39	0.9282	2.73	4.17	30.62	0.7666	6.96	14.08	29.38	0.7150	8.40	17.48
RCTV	37.79	0.9665	2.00	2.81	33.33	0.9075	5.50	**5.45**	31.43	0.8546	7.70	8.61
CTV	38.39	0.9675	1.94	2.62	34.83	0.9113	**4.02**	5.70	32.01	0.8320	**6.22**	10.62
LRTFR	35.02	0.9285	2.72	3.31	31.87	0.8642	9.17	8.75	30.72	0.8451	10.93	10.52
CRNL	34.70	0.9357	2.79	3.35	28.00	0.8100	8.94	11.42	26.26	0.7582	11.28	15.29
HIRDiff	34.43	0.9052	2.76	3.65	26.72	0.7538	9.74	12.91	24.41	0.6498	12.65	17.01
DTucker(PAN)	36.82	0.9573	2.21	3.06	31.83	0.8854	4.20	6.00	29.28	0.8285	6.24	10.58
PWRCTV(PAN)	39.34	0.9726	1.64	2.31	34.86	0.9170	5.16	5.87	32.24	0.8514	7.59	10.24
CCTD(PAN)	39.06	0.9715	1.81	2.54	**35.14**	**0.9340**	7.83	7.26	**34.14**	**0.9216**	8.94	**8.45**

Algorithm	Case 4				Case 5				Case 6			
	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓	PSNR↑	SSIM↑	ERGAS↓	SAM↓
NGMeet	22.21	0.5076	19.71	25.15	21.02	0.4386	22.04	29.24	23.74	0.5882	18.43	22.36
BALMF	27.52	0.6919	15.33	26.13	23.40	0.5322	23.57	41.18	27.99	0.7630	15.65	23.24
LRTV	26.10	0.6325	12.19	21.28	25.15	0.5649	14.59	28.51	28.28	0.7297	9.02	14.69
LMHTV	27.65	0.7344	11.77	20.28	26.24	0.6433	14.14	28.10	30.46	0.8392	10.57	15.20
LTHTV	27.33	0.7169	12.25	19.40	25.70	0.6274	16.27	28.66	29.61	0.8084	12.85	16.75
TCTV	28.28	0.6719	9.73	20.53	27.26	0.6338	**10.97**	23.36	28.89	0.7211	8.23	16.71
RCTV	28.91	0.7565	10.60	16.32	26.61	0.6395	13.69	27.08	32.27	0.8721	7.13	9.56
CTV	30.11	0.7557	**8.76**	16.65	28.67	0.6880	11.14	22.19	32.65	0.8543	**5.98**	10.09
LRTFR	29.75	0.8198	12.97	13.05	28.08	0.7862	16.37	16.70	30.89	0.8380	10.83	11.50
CRNL	24.89	0.6936	13.52	19.27	23.96	0.6713	15.38	22.95	26.15	0.7501	12.40	16.76
HIRDiff	22.22	0.5225	16.85	22.83	21.20	0.4609	19.15	26.43	22.90	0.5458	15.17	21.18
DTucker(PAN)	27.10	0.7333	11.77	16.90	25.82	0.6649	16.30	23.80	28.96	0.8094	9.47	13.09
PWRCTV(PAN)	29.52	0.7616	10.35	16.43	27.80	0.6672	14.60	26.37	32.62	0.8630	7.58	11.25
CCTD(PAN)	**32.91**	**0.9036**	10.42	**10.19**	**31.38**	**0.8741**	12.15	**12.68**	**34.29**	**0.9280**	7.97	**8.57**

TABLE IV

RESULTS OF DIFFERENT LOSS FUNCTIONS. BOLD INDICATES THE BEST VALUE

	Method	PSNR↑	SSIM↑	ERGAS↓	SAM↓
Case 1	PWRCTV	38.83	0.9755	**1.30**	**1.82**
	CCTD-SCG	39.11	0.9757	1.37	2.16
	CCTD	**39.25**	**0.9765**	1.34	2.11
Case 2	PWRCTV	33.71	0.9318	3.15	6.43
	CCTD-SCG	36.08	0.9707	1.81	3.15
	CCTD	**36.36**	**0.9723**	**1.76**	**3.05**
Case 3	PWRCTV	30.58	0.8654	5.52	10.67
	CCTD-SCG	34.97	0.9633	2.04	3.57
	CCTD	**35.20**	**0.9648**	**2.00**	**3.48**
Case 4	PWRCTV	27.96	0.7793	8.79	16.46
	CCTD-SCG	33.78	0.9536	2.37	4.20
	CCTD	**33.93**	**0.9550**	**2.35**	**4.13**
Case 5	PWRCTV	25.84	0.6784	11.97	23.03
	CCTD-SCG	32.12	0.9381	3.06	5.52
	CCTD	**32.23**	**0.9397**	**3.05**	**5.48**
Case 6	PWRCTV	29.95	0.8359	6.59	14.53
	CCTD-SCG	35.38	0.9677	2.04	3.79
	CCTD	**35.54**	**0.9686**	**2.02**	**3.78**

that cross-modal spectral constraints may introduce severe spectral distortion due to the large spectral difference between HSI and guidance modality.

TABLE V

RESULTS OF DIFFERENT MODELING STRATEGIES. BOLD INDICATES THE BEST VALUE

	$\mathcal{L}_{\text{HSI}}$	$\mathcal{L}_{\text{G}}$	$\mathcal{L}_{\text{LR}}$	Coupled	PSNR↑	SSIM↑	ERGAS↓	SAM↓
(a)	✓	×	×	✓	29.29	0.8317	7.99	18.67
(b)	✓	✓	×	✓	31.85	0.9260	4.49	11.90
(c)	✓	×	✓	✓	34.83	0.9729	1.55	4.37
(d)	✓	✓	✓	×	35.19	0.9815	1.38	4.03
(e)	✓	✓	✓	✓	**35.62**	**0.9836**	**1.32**	**3.88**

TABLE VI

RESULTS OF SINGLE-CHANNEL GUIDANCE ON WUHAN DATASET. BOLD INDICATES THE BEST VALUE

	Strategy	PSNR↑	SSIM↑	ERGAS↓	SAM↓
(a)	tensor nuclear norm	34.16	0.9808	1.48	3.91
(b)	spectral coupling	34.35	0.9789	2.03	5.29
(c)	CCTD	**35.62**	**0.9836**	**1.32**	**3.88**

F. Robustness Analysis

In the proposed CCTD framework, the guidance modalities are assumed to be clean. However, this assumption may not always hold in real scenarios, since guidance modalities acquired by different sensors can also be affected by noise. To evaluate the robustness of CCTD to noisy guidance information, different levels of Gaussian noise are added to the guidance modality to simulate realistic noise degradation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

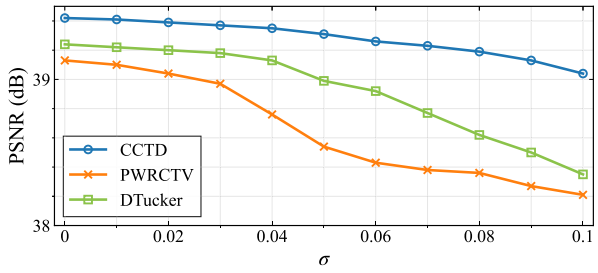


Fig. 6. Denoising visual comparisons on Florence dataset under noise case 3 (Bands: 61-16-48).

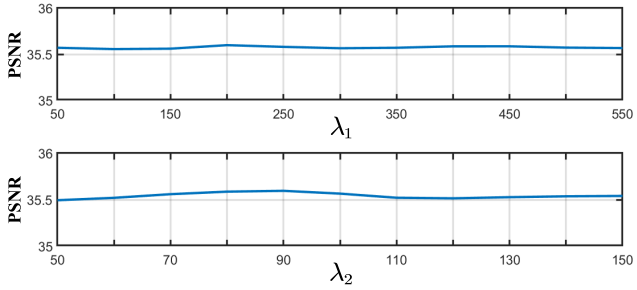


Fig. 7. PSNR values of CCTD with different settings of hyperparameters (the guidance weight λ_1, and the low-rank regularization weight λ_2) for HSI denoising on the Beijing dataset under noise Case 6.

Fig. 4 presents the PSNR robustness comparison of CCTD, PWRCTV, and DTucker as the noise level of the guidance modality increases. It can be observed that CCTD maintains a more stable performance than the compared multimodal guided denoising methods under increasing guidance noise. This demonstrates that the proposed CCTD framework is more robust to noisy guidance modalities.

G. Parameter Sensitivity Analysis

This section evaluates the parameter sensitivity of the CCTD method on the Beijing dataset under the complex noise conditions of Case 6. CCTD has only two primary hyperparameters: the guidance weight λ_1 and the low-rank regularization weight λ_2, which respectively govern the strength of the multimodal structural guidance and the spectral–spatial low-rank prior. As illustrated in Fig. 7, the CCTD framework maintains high PSNR values across a broad range of parameter settings and exhibits minimal sensitivity to minor fluctuations in hyperparameters. This stability confirms the practical applicability of the model and ensures it is straightforward to configure for diverse remote sensing restoration scenarios.

V. Discussion and Limitations

The CCTD framework provides a unified solution for guided HSI denoising with diverse modalities, thanks to its shared continuous representation. However, there are limitations: the MLP-based representation requires iterative optimization, which is slower than feed-forward deep learning methods (training takes 10 min per image). Future work could explore accelerating the optimization or learning a prior over the continuous representation. In addition, the method assumes perfect coregistration; slight misalignment between HSI and guidance could degrade performance—a direction for future robustness studies.

VI. Conclusion

This article proposed a unified guided hyperspectral image denoising framework, termed Continuous Coupled Tucker Decomposition (CCTD), that accommodates heterogeneous guidance modalities, including PAN, MSI, and SAR, within a common formulation. By combining coupled Tucker decomposition with continuous implicit neural representations, the proposed method jointly exploits cross-modal structural consistency and intrinsic spatial smoothness while preserving modalityspecific spectral characteristics. An additional nuclear norm regularization further reduces the sensitivity to Tucker rank selection. Experimental results on three benchmark datasets demonstrate that CCTD consistently outperforms state-of-the-art single-image and guided denoising methods under various noise conditions and guidance modalities.

References

[1] X. Cao, F. Zhou, L. Xu, D. Meng, Z. Xu, and J. Paisley, "Hyperspectral image classification with Markov random fields and a convolutional neural network," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2354–2367, May 2018.

[2] J. Wang, Z. Li, J. Yang, S. Liu, J. Zhang, and S. Li, "A multilevel spatial and spectral feature extraction network for marine oil spill monitoring using airborne hyperspectral image," *Remote Sens.*, vol. 15, no. 5, p. 1302, Feb. 2023.

[3] C. Zhang et al., "Medical hyperspectral image classification based weakly supervised single-image global learning network," *Eng. Appl. Artif. Intell.*, vol. 133, Jul. 2024, Art. no. 108042.

[4] Y. Chen, W. Lai, W. He, X.-L. Zhao, and J. Zeng, "Hyperspectral compressive snapshot reconstruction via coupled low-rank subspace representation and self-supervised deep network," *IEEE Trans. Image Process.*, vol. 33, pp. 926–941, 2024.

[5] S. Xu, C. Yu, J. Peng, S. Chen, X. Cao, and D. Meng, "Haar nuclear norms with applications to remote sensing imagery restoration," *IEEE Trans. Image Process.*, vol. 34, pp. 6879–6894, 2025.

[6] G. Chen and S.-E. Qian, "Denoising of hyperspectral imagery using principal component analysis and wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 3, pp. 973–980, Mar. 2011.

[7] B. Rasti, J. R. Sveinsson, M. O. Ulfarsson, and J. A. Benediktsson, "Hyperspectral image denoising using 3D wavelets," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 1349–1352.

[8] J. Xue, Y. Zhao, S. Huang, W. Liao, J. C.-W. Chan, and S. G. Kong, "Multilayer sparsity-based tensor decomposition for low-rank tensor completion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6916–6930, Nov. 2022.

[9] W. He, H. Zhang, L. Zhang, and H. Shen, "Hyperspectral image denoising via noise-adjusted iterative low-rank matrix approximation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3050–3061, Jun. 2015.

[10] J. Xue, Y. Zhao, Y. Bu, J. C.-W. Chan, and S. G. Kong, "When Laplacian scale mixture meets three-layer transform: A parametric tensor sparsity for tensor completion," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 13887–13901, Dec. 2022.

[11] H. Fan, C. Li, Y. Guo, G. Kuang, and J. Ma, "Spatial–spectral total variation regularized low-rank tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6196–6213, Oct. 2018.

[12] W. He, H. Zhang, L. Zhang, and H. Shen, "Total-variation-regularized low-rank matrix factorization for hyperspectral image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 178–188, Jan. 2016.

[13] L. Ding, Q. Wang, Y. Poo, and X. Zhang, "Nonlocal Gaussian scale mixture modeling for hyperspectral image denoising," *Comput. Vis. Image Understand.*, vol. 251, Feb. 2025, Art. no. 104270.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.

[14] N. Yokoya, T. Yairi, and A. Iwasaki, "Hyperspectral, multispectral, and panchromatic data fusion based on coupled non-negative matrix factorization," in *Proc. 3rd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2011, pp. 1–4.

[15] W. Liu, C. He, and L. Sun, "Spectral-smoothness and non-local self-similarity regularized subspace low-rank learning method for hyperspectral mixed denoising," *Remote Sens.*, vol. 13, no. 16, p. 3196, Aug. 2021.

[16] Z. Wang, M. K. Ng, L. Zhuang, L. Gao, and B. Zhang, "Nonlocal self-similarity-based hyperspectral remote sensing image denoising with 3-D convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531617.

[17] J. Xue, Y.-Q. Zhao, T. Wu, and J. C.-W. Chan, "Tensor convolution-like low-rank dictionary for high-dimensional image representation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 13257–13270, Dec. 2024.

[18] S. Xu, Q. Ke, J. Peng, X. Cao, and Z. Zhao, "Pan-denoising: Guided hyperspectral image denoising via weighted represent coefficient total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528714.

[19] Y. Chen, C. Ma, H. V. Poor, and Y. Chen, "Learning mixtures of low-rank models," *IEEE Trans. Inf. Theory*, vol. 67, no. 7, pp. 4613–4636, Jul. 2021.

[20] J. Bourcier, G. Dashyan, J. Chanussot, and K. Alahari, "Evaluating the label efficiency of contrastive self-supervised learning for multi-resolution satellite imagery," *Proc. SPIE*, vol. 12267, pp. 152–161, Oct. 2022.

[21] G. Perantoni and L. Bruzzone, "A novel technique for robust training of deep networks with multisource weak labeled remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[22] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled non-negative matrix factorization (CNMF) for hyperspectral and multispectral data fusion: Application to pasture classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2011, pp. 1779–1782.

[23] S. Xu et al., "Hipandas: Hyperspectral image joint denoising and super-resolution by image fusion with the panchromatic image," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 12002–12011.

[24] X. Huo and H. Gan, "Leveraging modal interaction and window dilation in attention network for hyperspectral and multispectral remote sensing image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5401715.

[25] D. M. McKeown, S. D. Cochran, S. J. Ford, J. C. McGlone, J. A. Shufelt, and D. A. Yocum, "Fusion of HYDICE hyperspectral data with panchromatic imagery for cartographic feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 3, pp. 1261–1277, May 1999.

[26] X. Lu, J. Zhang, T. Li, and Y. Zhang, "A novel synergetic classification approach for hyperspectral and panchromatic images based on self-learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4917–4928, Aug. 2016.

[27] X. Lu, J. Zhang, T. Li, and Y. Zhang, "Incorporating diversity into self-learning for synergetic classification of hyperspectral and panchromatic images," *Remote Sens.*, vol. 8, no. 10, p. 804, 2016.

[28] Y. Zhang, J. Chen, J. Wang, D. Shi, S. Han, and L. Deng, "C2DFF-net for object detection in multimodal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5407616.

[29] V. Sitzmann, J. Martel, A. Bergman, D. Lindell, and G. Wetzstein, "Implicit neural representations with periodic activation functions," in *Proc. NIPS*, 2020, pp. 7462–7473.

[30] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," *Commun. ACM*, vol. 65, no. 1, pp. 99–106, Dec. 2021.

[31] S. Xu, X. Cao, J. Peng, Q. Ke, C. Ma, and D. Meng, "Hyperspectral image denoising by asymmetric noise modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5545214.

[32] S. Xu et al., "Parameterized low-rank regularizer for high-dimensional visual data," *Int. J. Comput. Vis.*, vol. 133, no. 12, pp. 8546–8569, Dec. 2025.

[33] C. Peng et al., "Hyperspectral image denoising using nonconvex local low-rank and sparse separation with spatial–spectral total variation regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538617.

[34] Z. Xiao, "Hyperspectral image denoising based on low-rank matrix decomposition image prior," in *Proc. 6th Int. Conf. Electron. Commun., Netw. Comput. Technol. (ECNCT)*, Jul. 2024, pp. 56–59.

[35] L. Zhuang, X. Fu, M. K. Ng, and J. M. Bioucas-Dias, "Hyperspectral image denoising based on global and nonlocal low-rank factorizations," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10438–10454, Dec. 2021.

[36] M. Xie, X. Liu, and X. Yang, "Novel hybrid low-rank tensor approximation for hyperspectral image mixed denoising based on global-guided-nonlocal prior mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541817.

[37] H. Wang, J. Peng, W. Qin, J. Wang, and D. Meng, "Guaranteed tensor recovery fused low-rankness and smoothness," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10990–11007, Sep. 2023.

[38] J. Peng, Y. Wang, H. Zhang, J. Wang, and D. Meng, "Exact decomposition of joint low rankness and local smoothness plus sparse matrices," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5766–5781, May 2023.

[39] J. Peng, H. Wang, X. Cao, X. Liu, X. Rui, and D. Meng, "Fast noise removal in hyperspectral images via representative coefficient total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5546017.

[40] M. Wang et al., "Cross-satellite hierarchical multimodal denoising for hyperspectral imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 232, pp. 94–108, Feb. 2026.

[41] L. R. Tucker, "Some mathematical notes on three-mode factor analysis," *Psychometrika*, vol. 31, no. 3, pp. 279–311, Sep. 1966.

[42] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.

[43] Y. Liu et al., "Cross-city semantic segmentation (C2Seg) in multimodal remote sensing: Outcome of the 2023 IEEE WHISPERS C2Seg challenge," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8851–8862, 2024.

[44] W. He et al., "Non-local meets global: An iterative paradigm for hyperspectral image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2089–2107, Apr. 2022.

[45] S. Xu, J. Zhang, and C. Zhang, "Hyperspectral image denoising by low-rank models with hyper-Laplacian total variation prior," *Signal Process.*, vol. 201, Dec. 2022, Art. no. 108733.

[46] L. Pang, X. Rui, L. Cui, H. Wang, D. Meng, and X. Cao, "HIR-diff: Unsupervised hyperspectral image restoration via improved diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3005–3014.

[47] Y. Luo, X. Zhao, Z. Li, M. K. Ng, and D. Meng, "Low-rank tensor function representation for multi-dimensional data recovery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 3351–3369, May 2024.

[48] Y. Luo, X. Zhao, and D. Meng, "Revisiting nonlocal self-similarity from continuous representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 1, pp. 450–468, Jan. 2025.

[49] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[50] L. Wald, "Quality of high resolution synthesised images: Is there a simple criterion?," *Proc. 3rd Conf., Fusion Earth Data, Merging Point Meas., Raster Maps Remotely Sensed Images*, Sophia Antipolis, France, Jan. 2000, pp. 99–103.

[51] F. Kruse et al., "The spectral image processing system (SIPS)–interactive visualization and analysis of imaging spectrometer data," *Remote Sens. Environ.*, vol. 44, no. 2, pp. 145–163, 1993.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:47:40 UTC from IEEE Xplore. Restrictions apply.